

STOR 155.010 Lecture 12

Expectation and Variance

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Expectation, Variance, and Standard Deviation

Suppose X is a discrete RV taking values in $\{x_1, \dots, x_K\} \subset \mathbb{R}$.

- **Expected value/mean:** weighted average of its possible outcomes.

$$\mu = \mathbb{E}[X] = \sum_{i=1}^K x_i \mathbb{P}(X = x_i).$$

- **Variance:** measures variability in realized values

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_{i=1}^K (x_i - \mathbb{E}[X])^2 \cdot \mathbb{P}(X = x_i).$$

- **Standard Deviation:**

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

Example

- A basketball player's number of made three-pointers per game, X , has historically followed this distribution:

x	0	1	2	3	4
$\mathbb{P}(X = x)$	0.30	0.35	0.20	0.10	0.05

- What is the player's expected number of made three-pointers per game, $\mathbb{E}[X]$?
- What is the variance, $\text{Var}(X)$, and standard deviation, $SD(X)$?

Solution

x	$f_X(x)$	$x f_X(x)$	$(x - \mathbb{E}[X])^2$	$f_X(x) (x - \mathbb{E}[X])^2$
0	0.30	$0 \cdot 0.30 = 0$	$(0 - 1.25)^2 = 1.5625$	$0.30 \cdot 1.5625 = 0.4688$
1	0.35	$1 \cdot 0.35 = 0.35$	$(1 - 1.25)^2 = 0.0625$	$0.35 \cdot 0.0625 = 0.0219$
2	0.20	$2 \cdot 0.20 = 0.40$	$(2 - 1.25)^2 = 0.5625$	$0.20 \cdot 0.5625 = 0.1125$
3	0.10	$3 \cdot 0.10 = 0.30$	$(3 - 1.25)^2 = 3.0625$	$0.10 \cdot 3.0625 = 0.3063$
4	0.05	$4 \cdot 0.05 = 0.20$	$(4 - 1.25)^2 = 7.5625$	$0.05 \cdot 7.5625 = 0.3781$
		$\mathbb{E}[X] = 1.25$		$\text{Var}(X) = 1.2875$ $SD(X) = \sqrt{1.2875} \approx 1.13$

Combining Random Quantities

- Suppose, on average, each statistics homework problem takes you 10 minutes, and each chemistry problem takes 15 minutes.
- If you have 5 stats problems and 4 chemistry problems this week, what's your expected total time?

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- Suppose, on average, each statistics homework problem takes you 10 minutes, and each chemistry problem takes 15 minutes.
- If you have 5 stats problems and 4 chemistry problems this week, what's your expected total time?
- Intuitively: $5 \times 10 + 4 \times 15 = 50 + 60 = 110$ minutes.
- This “obvious” calculation (scaling averages and adding them up) is exactly what **linearity of expectation** formalizes.
- This works **no matter how the variables are related** to each other (independent or not).

Linear Combinations

- For random variables $X_1, \dots, X_k$ and constants $a_1, \dots, a_k$:

$$\mathbb{E}[a_1X_1 + \dots + a_kX_k] = a_1\mathbb{E}[X_1] + \dots + a_k\mathbb{E}[X_k].$$

In particular, $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. This always holds; no independence required.

- If $X_1, \dots, X_k$ are **independent**, then

$$\text{Var}(a_1X_1 + \dots + a_kX_k) = a_1^2 \text{Var}(X_1) + \dots + a_k^2 \text{Var}(X_k)$$

$$SD(a_1X_1 + \dots + a_kX_k) = \sqrt{\text{Var}(a_1X_1 + \dots + a_kX_k)}$$

In particular, if X and Y are **independent**, then

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$$

$$SD(aX + bY) = \sqrt{\text{Var}(aX + bY)}$$

Example 1

- Let X = number of two-point baskets made per game
 Y = number of three-point baskets made per game.
- Let $\mathbb{E}[X] = 6$ and $\mathbb{E}[Y] = 1.25$.
- A player's total points from field goals is

$$\text{Points} = 2X + 3Y$$

- Expected total points:

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- Expected total points:

$$\mathbb{E}[\text{Points}] = 2\mathbb{E}[X] + 3\mathbb{E}[Y] = 2(6) + 3(1.25) = 12 + 3.75 = 15.75$$

Linear Combinations with a Constant

- For a random variable X and constants a, c :

$$\begin{aligned}\mathbb{E}[aX + c] &= a\mathbb{E}[X] + c \\ \text{Var}(aX + c) &= a^2 \text{Var}(X)\end{aligned}$$

- A constant **shifts the mean**, but does not change the variance.
- In particular,

$$E(X + c) = \mathbb{E}[X] + c \quad \text{but} \quad \text{Var}(X + c) = \text{Var}(X).$$

- $\text{Var}(-X) = (-1)^2 \text{Var}(X) = \text{Var}(X)$. Flipping the sign of a random variable doesn't change its variance, since variance only measures spread, not direction.

Example 2

- A food truck sells burgers and tacos. Daily burger sales, B , has $\mathbb{E}[B] = 45$, $SD(B) = 6$. Daily taco sales, T , **independent** of B , has $\mathbb{E}[T] = 30$, $SD(T) = 4$.
- Burgers sell for \$8, tacos for \$5, and the truck pays a fixed \$50 daily rental fee.
- Daily net revenue: $R = 8B + 5T - 50$.
- Find $\mathbb{E}[R]$ and $\text{Var}(R)$.

- Expected value:

$$\mathbb{E}[R] = 8 \mathbb{E}[B] + 5 \mathbb{E}[T] - 50 = 8(45) + 5(30) - 50 = \$460$$

- Variance (B, T independent, so variances add):

$$\text{Var}(R) = 8^2 \text{Var}(B) + 5^2 \text{Var}(T) = 64(36) + 25(16) = 2304 + 400 = 2704$$

$$SD(R) = \sqrt{2704} = \$52$$

Example 3

- A store sells shirts (profit \$8 each) and returns pants (loss \$5 each).
- Let X = number of shirts sold per day, Y = number of pants returned per day; **independent** of each other. Let $\text{Var}(X) = 4$, $\text{Var}(Y) = 3$.
- Daily net profit:

Example 3

- A store sells shirts (profit \$8 each) and returns pants (loss \$5 each).
- Let X = number of shirts sold per day, Y = number of pants returned per day; **independent** of each other. Let $\text{Var}(X) = 4$, $\text{Var}(Y) = 3$.
- Daily net profit: $P = 8X - 5Y$.

$$\text{Var}(P) = 8^2 \text{Var}(X) + (-5)^2 \text{Var}(Y) = 64(4) + 25(3) = 256 + 75 = 331$$

$$SD(P) = \sqrt{331} \approx 18.19$$

- **Note:** the coefficients get squared.

Practice 1

- Ryan is a musician who works both as a solo artist and with his band. His earnings per month from solo gigs, S , has $E(S) = \$400$ and $SD(S) = \$100$. His earnings from band gigs, B , are **independent** of S , with $\mathbb{E}[B] = \$800$ and $SD(B) = \$200$.
- Find the expected value and standard deviation of Ryan's total monthly earnings, $T = S + B$.

Practice 2

- A tailor buys fabric for \$4 per yard and sells finished garments for \$25 each. Each garment uses 3 yards of fabric.
- Let N = number of garments sold in a month, with $\mathbb{E}[N] = 40$ and $SD(N) = 6$.
- Find the expected value and standard deviation of the monthly profit.

Practice 3

- A charity runs a raffle and a bake sale on the same day, and the two events attract completely different, unrelated crowds.
- Raffle revenue, R : $\mathbb{E}[R] = \$600$, $SD(R) = \$150$.
- Bake sale revenue, K : $\mathbb{E}[K] = \$250$, $SD(K) = \$50$.
- Fixed costs for the day (tables, permits, supplies) total \$180, paid regardless of turnout.
- Let net profit be $P = R + K - 180$. Find $\mathbb{E}[P]$ and $SD(P)$.

Practice 4

- A pizza shop tracks orders from two **independent** delivery apps. Orders via App A per day, X , has $\mathbb{E}[X] = 30$, $SD(X) = 5$. Orders via App B, Y , has $\mathbb{E}[Y] = 20$, $SD(Y) = 4$.
- The shop earns a \$6 profit per App A order, and a \$4 profit per App B order (App B charges a higher commission).
- Find the expected value and standard deviation of the total profit.